

Kunal Mishra, Ph.D.

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BIO

Computational biologist with expertise in systems genetics and multi-omics integration, spanning bulk, single-cell, and spatial transcriptomics. I lead kidney-focused research from computational discovery to experimental validation, with a focus on identifying causal genes and pathways in chronic kidney disease.

SKILLS

- **Computational and Programming:** R, Python, Bash, Snakemake, Git; reproducible workflow development, statistical modelling, data visualisation, high-performance computing environments
- **Omics and Systems Biology:** Bulk, single-cell, and spatial transcriptomics; multi-omics integration; systems genetics, QTL mapping, genetic association analysis, metabolomics
- **Experimental:** Cell culture, qPCR, IF/IHC, RNA-seq library preparation (bulk and single-cell); mouse and rat models

EDUCATION

- **Doctor of Philosophy (Ph.D.) in Computational Biology** Aug 2021 - Nov 2025
Centre for Computational Biology, Duke-NUS Medical School Singapore
 - Analyzed and integrated multi-omics datasets to study the role of macrophages in chronic kidney disease (the subject of my thesis)
 - Performed all computational analyses and end-to-end wet lab validation of computational findings
- **Bachelor of Science (BSc. Hons.) in Biological Sciences and Medicinal Chemistry** Aug 2017 - Jul 2021
Nanyang Technological University Singapore
 - Graduated with Distinction, Cumulative GPA: 4.42 / 5.00
 - Relevant Coursework: Bioinformatics Methods I & II, Metabolomics and Lipidomics, Advanced Immunology, Molecular & Cell Biology I & II, Organic Chemistry I-III, Drug Discovery and Development

EXPERIENCE

- **Computational Research Fellow** Dec 2026 - Present
Behmoaras Lab, Centre for Computational Biology, Duke-NUS Medical School Singapore
 - Lead kidney-focused research projects integrating systems genetics, multi-omics analysis, and experimental validation to identify causal genes and pathways in chronic kidney disease
 - Develop and apply reproducible computational pipelines for bulk, single-cell, and spatial transcriptomic analyses
 - Communicate findings through internal and external presentations and through first-author and co-author manuscripts for peer-reviewed publication
- **Bioinformatics Research Intern** May 2020 - Aug 2021
*Prabhakar Lab, Genome Institute of Singapore, A*STAR* Singapore
 - Developed and improved the performance of a novel single-cell feature selection algorithm (DUBStepR, Nature Communications, 2021)
 - Analysed single-nuclei transcriptomes from human fetal brains (Simons Foundation Autism Research Initiative-funded project)
 - Analyzed pilot Rheumatoid Arthritis Single-Cell RNA Sequencing data

RESEARCH WORK

‡ DENOTES EQUAL CONTRIBUTION

Publications

- [1] Wong BH, **Mishra K**, Chin CF, Galam DLA, Tan BC, Behmoaras J, Chua AWC, Silver DL. **Mfsd2a is important for maintaining epidermal homeostasis** (2026). *PNAS*, 123(8). DOI: 10.1073/pnas.2531159123
- [2] Chen T[‡], Quinn Z[‡], **Mishra K**[‡], O'Connor EC[‡], Silver SE, Zhang Y, Valencia MJ, Mei Y, Behmoaras J, Ferreira LMR, Chatterjee P. **moPPI: De novo generation of motif-specific and functionally active peptide binders via discrete flow matching** (2026). *bioRxiv*. doi:10.1101/2024.07.31.606098
- [3] Yen YH, Chew G, Chong CM, Nithianandam V, Kang PJ, Wee H, Yuan F, **Mishra K**, Behmoaras J, Cheung C, Zeng L, Su H, Ng ASL, Feany MB, Zhang SC. **FLT1-ICD loss orchestrates endothelial senescence and vascular dysfunction in Alzheimer's disease** (2025). (In Review)

- [4] Ouyang JF[‡], **Mishra K[‡]**, Xie Y, Park H, Huang KY, Petretto E, Behmoaras J. **Systems-level identification of a matrisome-associated macrophage polarisation state in multi-organ fibrosis** (2023). *eLife*, 12:e85530. DOI: 10.7554/eLife.85530
- [5] Ranjan B, Sun W, Park J, **Mishra K**, Schmidt F, Xie R, Alipour F, Singhal V, Joanito I, Honardoost MA, Yong JMY, Koh ET, Leong KP, Rayan NA, Lim MGL, Prabhakar S. **DUBStepR is a scalable correlation-based feature selection method for accurately clustering single-cell data** (2021). *Nature Communications*, 12:5849. DOI: 10.1038/s41467-021-26085-2

Conference Presentations

On profibrotic SPP1⁺ macrophages in disease

- **European Alliance of Associations for Rheumatology (EULAR) Symposium** 2026
Poster Tour talk. London, United Kingdom
- **6th CSI-JSI-KAI-SgSI Joint Symposium** 2026
Poster. Seoul, South Korea
- **Human Cell Atlas (General Meeting)** 2025
Poster. Singapore
- **2nd Spatial Biology Congress: Asia** 2024
Poster. Singapore

Systems Genetics for Diabetic Nephropathy

- **American Society of Nephrology (Kidney Week)** 2025
Poster. Houston, USA
- **Advances in Cardio-Renal-Metabolic Diseases (Duke-NUS)** 2024
Oral Talk. Singapore

Re-purposing anti-cancer compounds as anti-malarials:

- **International Conference of Undergraduate Research** 2019
Oral Talk. Virtual
- **Australasian Conference for Undergraduate Research** 2019
Oral Talk. Newcastle, Australia

HONORS AND AWARDS

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- **Duke-NUS Achievement Prize – Outstanding PhD Student** 2026
Duke-NUS Medical School
◦ Awarded for outstanding thesis research in Computational Biology
 - **Centre for Technology and Development Grant (SGD 50,000)** 2026
Duke-NUS Medical School
◦ Title: *Design and optimization of lead peptide candidates for fibrotic diseases*
 - **Best Lightning Talk – Advances in Cardio-renal-metabolic Diseases** 2024
Duke-NUS Medical School
◦ Title: *Systems Genetics in Diabetic Nephropathy for Causal Gene Identification*
 - **Best Poster Award – Ph.D. Student Symposium** 2023
Duke-NUS Medical School
◦ Title: *Understanding the spatiotemporal activity and role of SPP1⁺ profibrotic macrophages in kidney disease*
 - **Khoo Pre-Doctoral Fellowship** 2021
Duke-NUS Medical School
◦ Scholarship funding 4 years of stipend support and tuition
 - **NISTH-Micron Responsible AI Grant (SGD 5,000)** 2019
Nanyang Technological University
◦ Title: *Development of an interactive device for the elderly for prediction of dementia*
 - **Best Oral Presentation – Australasian Conference of Undergraduate Research** 2019
The University of Newcastle
◦ Title: *Identification of novel inhibitors for the malaria parasite Plasmodium falciparum*
◦ Received a travel grant (SGD 1,000) from NTU to present research overseas
 - **Presidential Research Scholar** 2019
Nanyang Technological University
◦ Awarded for undergraduate research accomplishments with merit
◦ Title: *Identification of novel inhibitors for the malaria parasite Plasmodium falciparum*
 - **Mr. and Mrs. Kwok Chin Yan Award for Student Initiative (SGD 1,000)** 2019
Nanyang Technological University
◦ Awarded for the *Code For Good* initiative by the NTU Open Source Society
◦ Led as President (2019–2020)

MEDIA

- **SGH to lead \$5.8m research project to combat deadly autoimmune disease**

The Straits Times; Singapore General Hospital (SGH)

2024

 
- **Scanning scars, Duke-NUS scientists find a ‘scarcode’ common across organs**

Duke-NUS NewsHub

2023



ADDITIONAL INFORMATION

Languages: English (Proficient), Hindi (Proficient), Indonesian (Conversational), French (Elementary)
Interests: Music (Guitar, Classical Flute), Photography